Week 2

- Stein's paradox

$$Y = \theta + \xi, \qquad \xi_i \text{ iid } \mathcal{N}(0, \sigma^2), \quad \theta \in \mathbb{R}^d$$

$$\hat{\theta}_{naive} = Y$$

$B_k = \{\theta \in \mathbb{R}^d : \theta \text{ is } k\text{-sparse}\}$ is not translation invariant!

$B_k + a \neq B_k$

But $\mathbb{R}^d$ is translation-invariant

$\mathbb{R}^d + a = \mathbb{R}^d$.

Thm (Stein's paradox): $\hat{\theta}_{naive}$ is inadmissible, i.e. there exists $\hat{\theta}$ such that

$$\mathbb{E}\|\hat{\theta} - \theta\|_2^2 \leq \mathbb{E}\|\hat{\theta}_{naive} - \theta\|_2^2, \quad \forall \theta,$$

and the inequality is strict for some $\theta \in \mathbb{R}^d$.

We can choose $\hat{\theta}$ as the James-Stein estimator:

$$\hat{\theta}_{JS} := \left(1 - \frac{\sigma^2(d-2)}{\|Y\|^2}\right) Y$$

In general, consider estimator of the form

$$\hat{\theta} = g(Y)\,Y,$$

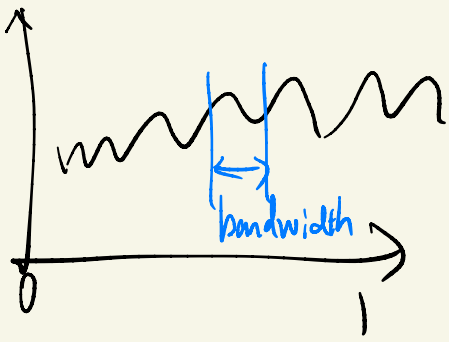
g: function $\mathbb{R}^d \to \mathbb{R}$

Stein's unbiased risk estimator (SURE): An unbiased estimator of the risk $\mathbb{E}\|\hat{\theta} - \theta\|_2^2$ is

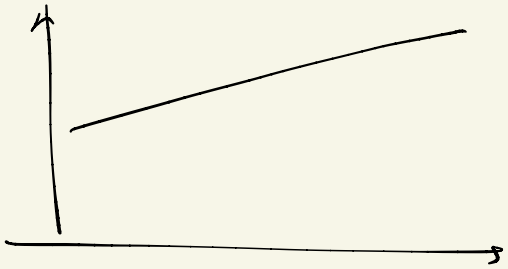
$$\text{SURE} = \sigma^2 d\,(2g(Y) - 1) + 2\sigma^2 \sum_{i=1}^{d} Y_i \frac{\partial g}{\partial y_i}(Y) + \|Y\|_2^2 (1 - g(Y))^2$$

SURE: function of Y (not on θ)

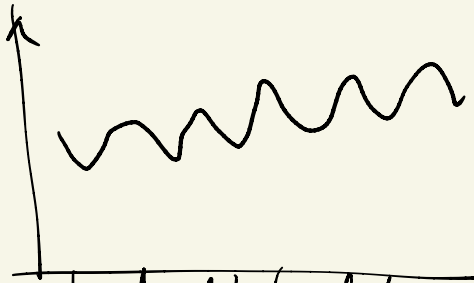
$$\mathbb{E}[\text{SURE}(Y)] = \mathbb{E}\|\hat{\theta} - \theta\|_2^2$$



large bandwidth



small bandwidth



Cross-validation → estimated risk as a function of the bandwidth (and the signal)

Proof of SURE:

$$\mathbb{E}\|\hat{\theta}-\theta\|_2^2 = \sum_{i=1}^{d} \mathbb{E}\left(g(Y)Y_i-\theta_i\right)^2$$

$$= \sum_{i=1}^{d}\Big\{\underbrace{\mathbb{E}(Y_i-\theta_i)^2}_{①} + \underbrace{2\mathbb{E}[(\theta_i-Y_i)(1-g(Y))Y_i]}_{②} + \mathbb{E}[Y_i^2(1-g(Y))^2]\Big\}$$

① $= \sigma^2$

② : use Gaussian integration by parts : if ϕ is function on $\mathbb{R}$, $X \sim N(0,1)$, then

$$\mathbb{E}[X\phi(X)] = \mathbb{E}[\phi'(X)]$$

$$\Rightarrow ② = -2\sigma^2\mathbb{E}\left[1-g(Y)-Y_i\frac{\partial g}{\partial y_i}(Y)\right]$$

Thm: If $d \geq 3$, $\theta \in \mathbb{R}^d$, then

$$\mathbb{E}\|\hat{\theta}_{JS}-\theta\|^2 = d\sigma^2 - \mathbb{E}\left[\frac{\sigma^4(d-2)^2}{\|Y\|^2}\right]$$

$$< d\sigma^2$$

$$= \mathbb{E}\|Y-\theta\|^2$$

Proof: By SURE, $\mathbb{E}\|\hat{\theta}-\theta\|^2$

$$= d\sigma^2 + \mathbb{E}[W(Y)]$$

where

$$W(y) := -2\sigma^2 d\,(1-g(y)) + 2\sigma^2 \sum_{i=1}^{d} y_i \frac{\partial g}{\partial y_i}(y) + \|y\|_2^2 \,(1-g(y))^2$$

Choose $g(y) = 1 - \frac{c}{\|y\|^2}$, $c > 0$

minimizing $W(y)$ gives $c_{opt} = \sigma^2(d-2)$ ∎

- Least squares regression.

$$Y = X\theta^* + \xi$$

$$X \in \mathbb{R}^{n \times d}, \quad d \le n, \quad \text{rank}(X) = d$$

$$\hat{\theta}^{LS} = \arg\min_{\theta} \|Y - X\theta\|_2^2$$

$$= (X^T X)^{-1} X^T Y$$

in general, Moore-Penrose pseudoinverse of $X^T X$.

Thm: $\mathbb{E}\,\|X(\hat{\theta}^{LS} - \theta)\|_2^2 = d\sigma^2$

Remark: If $\text{rank}(X) = r < d$, then $\mathbb{E}\,\|X(\hat{\theta}^{LS} - \theta)\|_2^2 = r\sigma^2$.

Proof: $\mathbb{E}\,\|X(\hat{\theta}^{LS} - \theta)\|_2^2$

$$= \mathbb{E}\,\|X(X^TX)^{-1}X^TY - X\theta\|_2^2$$

$$= \mathbb{E} \| X(X^TX)^{-1}X^T\xi \|_2^2$$

$$= \mathbb{E}\, \mathrm{Tr}\left[X(X^TX)^{-1}X^T\xi\xi^T X(X^TX)^{-1}X^T \right]$$

$$= \mathrm{Tr}\left[X(X^TX)^{-1}X^T \underbrace{\mathbb{E}[\xi\xi^T]}_{\sigma^2 I_d} X(X^TX)^{-1}X^T \right]$$

$$= \sigma^2 \mathrm{Tr}\left[X(X^TX)^{-1}X^T \right]$$

$$= \sigma^2 \mathrm{Tr}\left[X^TX(X^TX)^{-1} \right]$$

cyclic property: $\mathrm{Tr}[AB] = \mathrm{Tr}[BA]$

$$= \sigma^2 \mathrm{Tr}[I_d]$$

$$= d\sigma^2$$

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- Bayes Information Criterion (BIC) estimator

$$\hat{\theta}^{BIC} := \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \left\{ \| Y - X\theta \|_2^2 + \tau^2 \|\theta\|_0 \right\}$$

$\|\theta\|_0 = \sum_{i=1}^{d} 1_{\theta_i \neq 0}$

Likelihood: $e^{-\frac{\|Y-X\theta\|_2^2}{2}}$

prior density $p_\theta \propto e^{-\tau\|\theta\|_0}$

Note: when $X = I_d$, $n = d$, BIC $\Longleftrightarrow$ Hard thresholding

- Lasso :

$$\hat{\theta}^L = \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \left\{ \|Y - X\theta\|_2^2 + 2\tau \|\theta\|_1 \right\}$$

$$\|\theta\|_1 = \sum_{i=1}^{d} |\theta_i|$$

When $X = I_d$, $n = d$

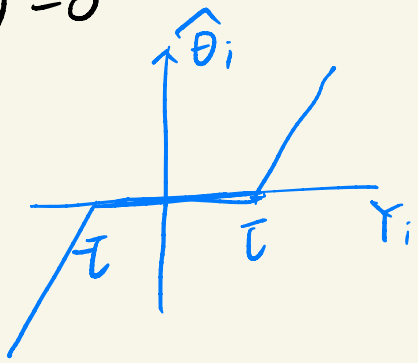
$$\Rightarrow \underset{\theta \in \mathbb{R}^d}{\operatorname{argmin}} \left\{ \|Y - \theta\|_2^2 + 2\tau \|\theta\|_1 \right\}$$

$$\Rightarrow \hat{\theta}_i^L = \underset{\theta}{\operatorname{argmin}} \left\{ (Y_i - \theta_i)^2 + 2\tau |\theta_i| \right\}$$

derivative $\Rightarrow -2(Y_i - \hat{\theta}_i) + 2\tau \operatorname{sign}(\hat{\theta}_i) = 0$

$$\hat{\theta}_i = Y_i - \tau \operatorname{sig}(\hat{\theta}_i)$$

soft-thresholding



- Computation : BIC is NP-hard !

  Lasso is convex opt. (glmnet in R)

- Fast rate for the Lasso

Definition : X satifies (k, κ)-restricted eigenvalue condition (RE) if $\inf_{|S| \le k} \inf_{\theta \in C_S} \frac{\|X\theta\|_2^2}{\|\theta_S\|_2^2} \ge \kappa$

where $C_S = \{\theta : \|\theta_{S^c}\|_1 \le 3\|\theta_S\|_1\}$.

$\theta_{S,i} = \begin{cases} \theta_i & \text{if } i \in S \\ 0 & \text{if } i \notin S \end{cases}$

Remark: typical behavior: RE ✓ for random design if $n \gtrsim k \log d$

Thm: Assume ground truth $\|\theta^*\|_0 \le k$, X satisfies $(k, 1/2)$-RE condition, each column $\|X_j\|_2^2 \le 2$. Let Lasso regularization parameter τ satisfy

$$2\tau = 8\sigma\left(\sqrt{\log(2d)} + \sqrt{\log\frac{1}{\delta}}\right).$$

Then

$$\|X(\hat{\theta}^L - \theta^*)\|_2^2 \lesssim k\sigma^2 \log(2d/\delta)$$

with probability $\ge 1-\delta$.

Notes: ① $f \lesssim g$ means there exists constant $a > 0$ such that $f \le ag$.

② $f \asymp g$, or $f \sim g$ if $f \lesssim g$ and $g \lesssim f$.

③ $\tau \asymp \sigma\left(\sqrt{\log d} + \sqrt{\log\frac{1}{\delta}}\right)$

$$\asymp \sigma\sqrt{\log d + \log\frac{1}{\delta}}$$

$$= \sigma\sqrt{\log\frac{d}{\delta}}$$

Proof: opt condition for Lasso:

$$\|Y - X\hat{\theta}^L\|_2^2 \le \|Y - X\theta^*\|_2 + 2\tau\|\theta^*\|_1 - 2\tau\|\hat{\theta}^L\|_1$$

($Y - X\hat{\theta}^L = X(\theta^* - \hat{\theta}^L) + \xi$; $Y - X\theta^* = \xi$)

$$\Rightarrow \|X(\hat{\theta}^L - \theta^*)\|_2^2 \le 2\xi^T X(\hat{\theta}^L - \theta^*) + 2\tau\|\theta^*\|_1 - 2\tau\|\hat{\theta}^L\|_1 \quad (1)$$

$$\xi^T X(\hat{\theta}^L - \theta^*) \overset{\text{Hölder's inequality}}{\le} \|\xi^T X\|_\infty \|\hat{\theta}^L - \theta^*\|_1$$

$$\le \frac{\tau}{2}\|\hat{\theta}^L - \theta^*\|_1 \quad (2)$$

→ $\mathrm{Var}(\xi^T X_j) \le \|X_j\|_2^2 \le 2$

use the fact: if $z_1 \cdots z_d$ are Gaussian (not independent) with var ≤ 2, then

$$\max_{1 \le i \le d} z_i \le \sqrt{\log d/\delta}$$

with probability $\ge 1 - \delta$

(proof by union bound)

(1) & (2) $\Rightarrow$ $\|X(\hat{\theta}^L - \theta^*)\|_2^2$

$$\le \tau\|\hat{\theta}^L - \theta^*\|_1 + 2\tau\|\theta^*\|_1 - 2\tau\|\hat{\theta}^L\|_1$$

(use $\theta^*_S = \theta^*$)

$$= \tau\|\hat{\theta}^L_S - \theta^*_S\|_1 - \tau\|\hat{\theta}^L_{S^c}\|_1 + 2\tau\|\theta^*_S\|_1 - 2\tau\|\hat{\theta}^L_S\|_1$$

triangle

$$\le 3\tau\|\hat{\theta}^L_S - \theta^*_S\|_1 - \tau\|\hat{\theta}^L_{S^c}\|_1 \quad (3)$$

In particular, $3\|\theta^L_S - \theta^*_S\|_1 \geq \|\hat{\theta}^L_{S^c}\|_1 = \|\hat{\theta}^L_{S^c} - \theta^*_{S^c}\|_1$

So

$$\|\hat{\theta}^L_S - \theta^*\|_1 \overset{\text{Cauchy-Schwarz}}{\leq} \sqrt{k}\,\|\hat{\theta}^L_S - \theta^*\|_2$$

$$\overset{\text{RE}}{\leq} \sqrt{2k}\,\|X(\hat{\theta}^L - \theta^*)\|_2$$

Plugging into (3)

$$\|X(\hat{\theta}^L - \theta^*\|_2^2 \leq 3\tau \cdot \sqrt{2k}\,\|X(\hat{\theta}^L - \theta^*)\|_2$$

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